Contents

Electrochemistry

Electrochemistry

Electrochemistry :-

It is the study of processes involved in the interconversion of electrical energy & chemical energy.

Ex:- Discharging of a battery

→ Chemical reactions takes place in a solution at the interface of an electron conductor (a metal or semi-conductor) & an ionic conductor (an electrolyte) & which involve electron transfer between the electrode and the electrolyte or species in solution.

→ The interconvesions of energies takes place through redox reactions (both oxidation & reduction) from the basis of electrochemical cells.

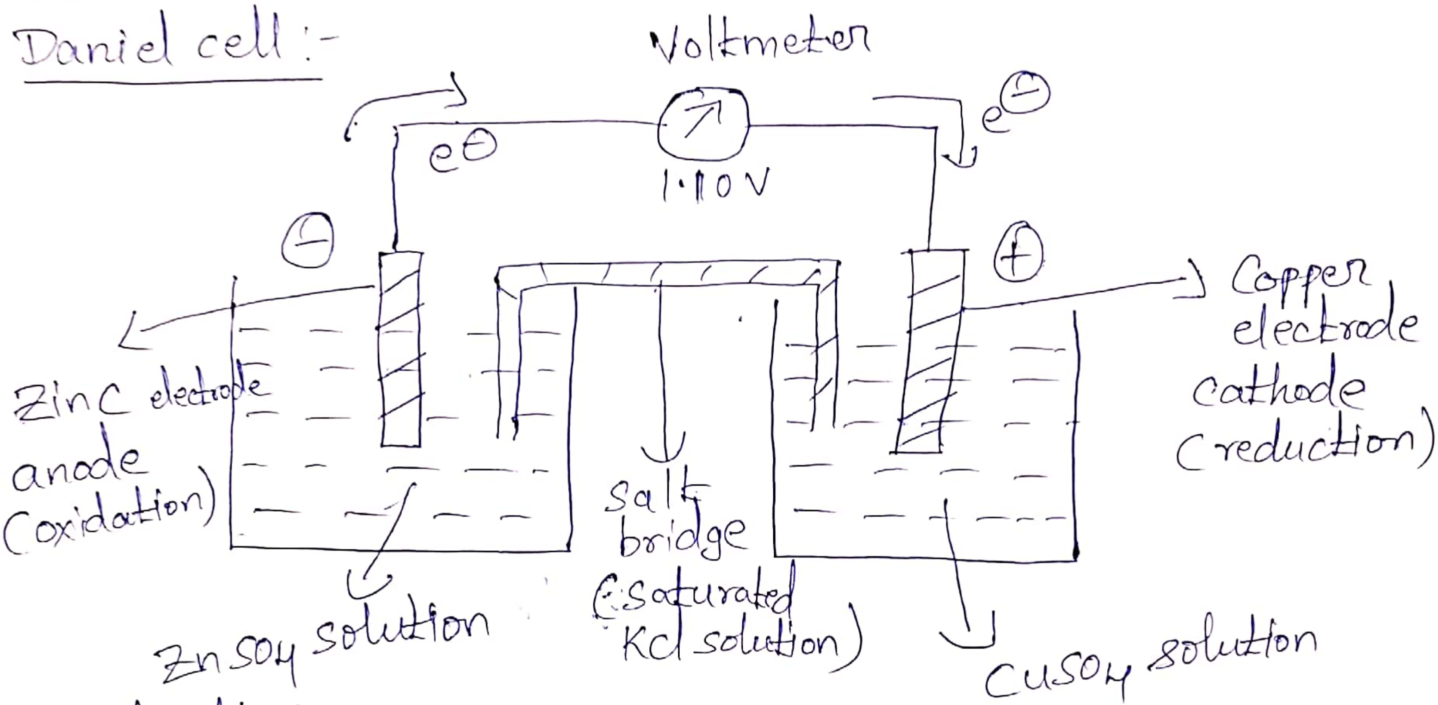
Electrochemical cells

- Electrolytic cells (conductance cells)
  - (converts electrical energy to chemical energy)
- Galvanic (or) voltaic cells (converts chemical energy to electrical energy)
  - Chemical cells
    - Primary (or) irreversible cells
    - Secondary (or) reversible cells
  - Concentration cells

Construction & working of Galvanic (or) voltaic cell :-

The 'Daniel cell' is a typical example for the 'Galvanic cell'.

Daniel cell :-



Construction :-

Daniel cell (Galvanic cell) consists of two different electrodes dipped in two electrolyte solutions which are ~~separated by~~ connected through a salt bridge. The zinc rod dipped in zinc sulphate solution & copper rod in ~~zinc~~ copper sulphate solution.

Working :-

Daniel cell can be represented as

$$Zn/ZnSO_4 \,||\, CuSO_4/Cu$$

Cell reactions :-

At anode (or) negative electrode (left electrode) :-

At the zinc electrode, ~~zi~~ oxidation takes place and zinc goes into the solution as Zn^{2+} ions liberating two electrons

$$Zn \longrightarrow Zn^{2+} + 2e^{\ominus} \text{ (oxidation)}$$

At cathode (or) positive electrode (right electrode):

At the copper electrode, copper reduction takes place & electrode reaction of copper ions (Cu^{2+}) takes place with the deposition of metallic copper on copper rod by consuming two electrons.

$$Cu^{2+} + 2e^{\ominus} \longrightarrow Cu \text{ (reduction)}$$

The net chemical cell reaction is:

$$Zn \longrightarrow Zn^{2+} + \cancel{2}e^{\ominus} \text{ (oxidation)}$$
$$Cu^{2+} + 2e^{\ominus} \longrightarrow Cu \text{ (reduction)}$$

Cell reaction: $Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$

oxidation reduction

For the above cell, the emf is found to be 1.1V at 25°C (298 K). Therefore, the emf of the cell is positive.

→ In Galvanic cell, emf can be expressed as the difference between two single electrode potential

The cell emf, $E_{cell} = E_{cathode} - E_{anode}$

—d—

Some important terms in Electrochemistry:-

1) Electromotive force (emf) of the cell (Ecell):-

"The potential difference between the two electrodes of a galvanic cell which causes the flow of current from one electrode (higher potential) to the other (lower potential) as a result of spontaneous redox reaction is called the emf of the cell (or) the cell potential."

"Emf of the cell" is denoted by "Ecell".

→ The emf of the cell depends on the nature of the electrode, temperature and concentration of the electrolyte solutions.

∴ Emf of the cell, $E_{cell} = E_{cathode} - E_{anode}$ (or) $E_{right} - E_{left}$

2) standard electrode potential (E^0_{cell}):-

The potential that is developed when an element (metal or non-metal) is in equilibrium with a solution containing its own ions of unit concentration (1M) at 25°C (298K) is called "standard electrode potential".

→ It's denoted by E^0_{cell}.

3) Single electrode potential (or) electrode potential (or) reduction potential (E (or) $E_{M^{n+}/M}$):-

It's defined as the potential developed at the interface between the metal and the solution, when it is in contact with a solution of its ions.

→ It's denoted by $E_{M^{n+}/M}$ (or) E

Ex:- Reduction potential of zinc electrode is

$E_{Zn^{2+}/Zn} = -0.76V$

Oxidation potential of zinc electrode is

$E_{Zn/Zn^{2+}} = 0.76V$

—✓—

Origin of single electrode potential:-

"The potential developed at the interface between the metal and the solution, when it is in contact with a solution of its ions".

→ whenever a metal is in contact with its own ions, it has a natural tendency to loose or gain electrons. Because of this tendency, the electrode gains a potential, which is known as 'single electrode potential'.

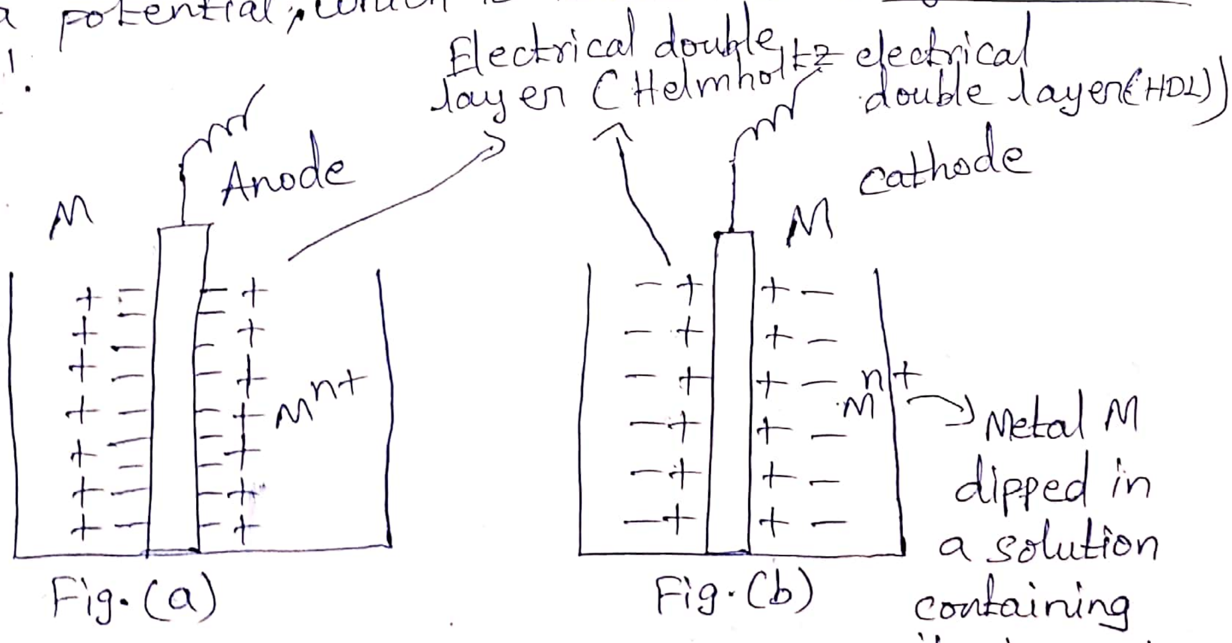


Figure.1

If a metal (M) dipped in a solution containing its ions (M^{n+}), the following two reactions are possible.

i) Metals go into solution as metal ions by loosing electrons. (Fig. 1(a)).

The electrons accumulate on the electrode surface making it negatively charged. The negatively charged electrode surface attracts a layer of positively charged ions (M^{n+}) at the interface developing an "electrical double layer" at the metal solution interface which is called "Helmholtz Electric double layer (HDL)" (Fig. 1(a)). The potential develops across this double layer & this potential is called single electrode potential.

$$M \longrightarrow M^{n+} + ne^{\ominus} \quad \text{(oxidation)}$$

loss of electrons

ii) At the same time, metal ions (M^{n+}) in the solution get deposited as metal atoms by consuming electrons. (Fig. 1(b)).

The positively charged metal ions accumulate on the electrode surface making it positively charged. The positively charged electrode surface attracts a layer of negatively charged ions at the interface & again developing an "electrical double layer" which is called "Helmholtz Electric double layer (HDL)" (fig. 1(b)). This double layer again develops a potential called single electrode potential.

$M^{n+} + ne^{\ominus} \longrightarrow M$ (reduction, gain of electrons)

— d —

** Nernst equation for a single electrode potential (thermodynamic concept of cell potential (or) electrode potential) :-

In 1889, Nernst derived a "quantitative relationship between electrode potential and concentration of the electrolyte species involved".
This expression is known as "Nernst equation".

→ Nernst equation relates single electrode potential (E) to its standard electrode potential (E^{o}), concentration of metal ions (M^{n+}) & temperature (T).

→ In order to derive this equation, free energy concept is used.

Consider the following reversible electrode reaction

$$M^{n+} + ne^{\ominus} \underset{}{\overset{K}{\rightleftharpoons}} M$$

The decrease in free energy is related to the maximum work done.

$$-\Delta G = W_{max} \quad \text{(1)}$$

Maximum work available from a reversible chemical process is equal to maximum amount of electrical energy that can be obtained.

W_{max} = Electrical energy (work) produced $= nFE$

Here, n is number of electrons involved in reaction

F = Faraday's constant = 96,500 coulombs

Hence, decrease in free energy is

$$\boxed{-\Delta G = nFE} \longrightarrow (1)$$

ΔG^0 = standard free energy change $= -nFE^0 \longrightarrow (2)$

Since, $K = \frac{[\text{product}]}{[\text{reactant}]} = \frac{[M]}{[M^{n+}]} \longrightarrow (3)$

K stands for reaction quotient of the activities of the products & reactants at any given stage of the reaction (or) equilibrium constant.

Thermodynamic relationship known as Van't Hoff's isotherm equation that relate free energy with equilibrium

$$\boxed{\Delta G = \Delta G^0 + RT \ln K_c} \longrightarrow (4)$$

Substituting equations (1), (2), & (3) in equation (4), we get

$$-nFE = -nFE^0 + RT \ln \frac{[M]}{[M^{n+}]}$$

Divide by $-nF$ throughout the above equation, & substitute ln = 2·303 log, & [M] = 1

[∵ concentration of pure metal is taken as unity]

$$E = E^{0} - \frac{2.303\,RT}{nF} \log \frac{1}{[M^{n+}]}$$

By rearranging the above equation, the Nernst equation for single electrode potential is obtained.

$$\boxed{E = E^{0} + \frac{2.303\,RT}{nF} \log [M^{n+}]}$$

Substituting gas constant $(R) = 8.314$ KJ/mol

Faraday's constant $(F) = 96{,}500$ coulombs/mol

Temperature $(T) = 298K$ (or) $25^{0}C$

$$\boxed{E = E^{0} + \frac{0.0591}{n} \log [M^{n+}]}$$ at 298K (25°C)

Nernst equation for a Galvanic cell :-

Nernst equation can be used to calculate emf of a galvanic cell. To do that, substitute Nernst equation for single electrode potential for anode & cathode in the following emf equation

$$E_{cell} = E_{cathode} - E_{anode}$$

$$E_{cell} = \left\{E^{0}_{cathode} + \frac{2.303\,RT}{nF} \log [M^{n+} \text{at cathode}]\right\} - \left\{E^{0}_{anode} + \frac{2.303\,RT}{nF} \log [M^{n+} \text{at anode}]\right\}$$

$$E_{cell} = (E^{0}_{cathode} - E^{0}_{anode}) + \frac{2.303\,RT}{nF} \log \frac{[M^{n+} \text{at cathode}]}{[M^{n+} \text{at anode}]}$$

$$\boxed{E_{cell} = E^{0}_{cell} + \frac{2.303\,RT}{nF} \log \frac{[M^{n+} \text{at cathode}]}{[M^{n+} \text{at anode}]}}$$

Scanned with CamScanner

$$\boxed{E_{cell} = E^o_{cell} + \frac{0.0591}{n} \log \frac{[M^{n+} \text{at cathode}]}{[M^{n+} \text{at anode}]}}$$

at 298K (25°c)

— 𝓁 —

***

Formulas for numerical problems:-

1) Galvanic (or) voltaic cell (Daniel cell):-

$$\boxed{\text{emf. } E_{cell} = E_{cathode} - E_{anode}}$$

Nernst equation for a Galvanic cell,

$$E_{cell} = E^0_{cell} + \frac{2.303RT}{nF} \log \frac{[M^{n+} \text{at cathode}]}{[M^{n+} \text{at anode}]}$$

(or)

$$E_{cell} = E^0_{cell} + \frac{0.0591}{n} \log \frac{[M^{n+} \text{at cathode}]}{[M^{n+} \text{at anode}]}$$

at 298K (25°c)

2) Nernst equation for a single electrode potential (or) reduction potential:-

$$E_{cell} = E^0_{cell} + \frac{2.303\,RT}{nF} \log [M^{n+}]$$

(or)

$$E_{cell} = E^0_{cell} + \frac{0.0591}{n} \log [M^{n+}]$$

at 298K (25°c)

3) Concentration cell :-

$$E_{cell} = \frac{2.303RT}{nF} \log \left[\frac{C_2}{C_1}\right]$$

(or)

$$E_{cell} = \frac{0.0591}{n} \log \left[\frac{C_2}{C_1}\right] \text{ at } 298K\,(25^\circ C)$$

$$\boxed{C_2 > C_1}$$

***

Numerical problems on Nernst equation for a Single electrode potential:

1) The standard electrode potential of Zn electrode is $-0.76\,V$ and the concentration $[Zn^{2+}]$ given $= 0.25M$. Calculate $E_{Zn^{2+}/Zn}$ at $28°c$.

↳ electrode potential of Zinc.

A) Nernst equation for single electrode potential at $28°c$ is

$$\boxed{E_{cell} = E^0_{cell} + \frac{2.303RT}{nF}\log[M^{n+}]}$$

$$E_{Zn^{2+}/Zn} = E^0_{Zn^{2+}/Zn} + \frac{2.303RT}{nF}\log[Zn^{2+}]$$

$$E_{Zn^{2+}/Zn} = -0.76V + \frac{2.303 \times 8.314\ KJ/Kg\ mol \times 301K}{2 \times 96,500\ coulombs/mol}\log[0.25]$$

$$\left[\because T = 28°c\ (given)\quad T = 273+28\right]$$

$$T = 301K$$

$$\boxed{E_{Zn^{2+}/Zn} = -0.78V}$$

---

2) calculate the electrode potential of zinc, when the standard electrode potential of zinc electrode is $-0.76V$ & concentration of Zn^{2+} ion is $0.1M$ at $25°c$.

A) $E^0_{Zn^{2+}/Zn} = -0.76V$

$[Zn^{2+}] = 0.1M$

Nernst equation for a single electrode potential at 25°c (298K) is

$$\boxed{E_{Zn^{2+}/Zn} = E^{0}_{Zn^{2+}/Zn} + \frac{0.0591}{n} \log [Zn^{2+}]}$$

$$E_{Zn^{2+}/Zn} = -0.76V + \frac{0.0591}{2} \log [0.1]$$

$$E_{Zn^{2+}/Zn} = -0.07895 \text{ V}$$

---

3) Calculate the $E^{0}_{Cu^{2+}/Cu}$, if the potential of Cu electrode immersed in 0.015 M Cu^{2+} solution is 0.296V at 25°c.

A) Nernst equation for single electrode potential at 25°C is

$$\boxed{E = E^{0} + \frac{0.0591}{n} \log [M^{n+}]}$$

Rearranging the above equation

$$E^{0}_{Cu^{2+}/Cu} = E_{Cu^{2+}/Cu} - \frac{0.0591}{n} \log [Cu^{2+}]$$

$$E^{0}_{Cu^{2+}/Cu} = 0.296 - \frac{0.0591}{2} \log [0.015]$$

$$E^{0}_{Cu^{2+}/Cu} = 0.349 \text{ V}$$

---

4) calculate the electrode potential of Fe at 25°c, when the electrode potential of $Fe^{2+}(0.1M)/Fe$ given is $E^{0}_{Fe^{2+}/Fe} = -0.44V$.

A) Nernst equation for single electrode potential at 25°C is

$$\boxed{E_{Fe^{2+}/Fe} = E^0_{Fe^{2+}/Fe} + \frac{0.0591}{n} \log [Fe^{2+}]}$$

$$E_{Fe^{2+}/Fe} = E^0_{Fe^{2+}/Fe} + \frac{0.0591}{2} \log [0.1]$$

$$E_{Fe^{2+}/Fe} = -0.44 + \frac{0.0591}{2} \log [0.1]$$

$$E_{Fe^{2+}/Fe} = -0.4695V.$$

5) Calculate the reduction potential (or) electrode potential of copper when it is in contact with 5M $CuSO_4$ solution at 298 K. The E^0 value of Cu electrode is 0.34V.

A) Nernst equation for single electrode potential at 25°C (298K) is

$$\boxed{E_{Cu^{2+}/Cu} = E^0_{Cu^{2+}/Cu} + \frac{0.0591}{n} \log [Cu^{2+}]}$$

$$E_{Cu^{2+}/Cu} = 0.34 + \frac{0.0591}{2} \log [5]$$

$$E_{Cu^{2+}/Cu} = 0.36 \text{ V}$$

— x —

***Numerical problems on emf of a cell (Galvanic (or) voltaic cell)

1) Calculate the emf of a cell containing Cu & Ag electrodes. Given that the electrode potentials of Cu & Ag are 0.34V & 0.8V respectively.

A) $$\boxed{E_{cell} = E^0_{cathode} - E_{anode}}$$

Cell representation is

$Cu / Cu^{2+}(1M) // Ag^{+}(1M) / Ag$

$E_{cell} = 0.8 - 0.34 = 0.46V$

Electrode reactions are :

At anode :- $Cu \rightarrow Cu^{2+} + 2e^{\ominus}$

At cathode :- $2Ag^{+} + 2e^{\ominus} \rightarrow 2Ag$

Cell reaction :- $Cu + 2Ag^{+} \rightarrow Cu^{2+} + 2Ag$

—x—

---

2) Write the electrode reactions, cell reactions & calculate the emf of the cell at 298K for the cell formed by Fe^{2+}/Fe, $E^{0} = -0.44$ & Zn^{2+}/Zn $E^{0} = -0.76V$.

A) Cell representation is

$Zn / Zn^{2+} // Fe^{2+} / Fe$

Electrode reactions are :

At anode : $Zn \rightarrow Zn^{2+} + 2e^{\ominus}$

At cathode :- $Fe^{2+} + 2e^{\ominus} \rightarrow Fe$

cell reaction : $Zn + Fe^{2+} \rightarrow Zn^{2+} + Fe$

∴ Emf of the cell

$E_{cell} = E_{cathode} - E_{anode}$

$E_{cell} = -0.44 - (-0.76)$

$E_{cell} = 0.32V$

3) A galvanic cell formed by immersing Cd rod in 1M $CdSO_4$ & Cu rod in 1M $CuSO_4$ solution. The standard E^0 values ~~of~~ of Cu & Cd electrodes are +0.34 V & −0.40 V.

a) How the cell is represented?

b) write the electrode reactions & cell reaction.

c) Calculate the standard emf of the cell at 298 K.

A) Cell representation is

$$Cd / Cd^{2+}(1M) // Cu^{2+}(1M) / Cu$$

Electrode reactions are

At anode: $Cd \longrightarrow Cd^{2+} + 2e^{\ominus}$

At cathode: $Cu^{2+} + 2e^{\ominus} \longrightarrow Cu$

Cell reaction: $Cd + Cu^{2+} \longrightarrow Cd^{2+} + Cu$

Emf of the cell $E_{cell} = E_{cathode} - E_{anode}$

$$E_{cell} = E_{Cu} - E_{Cd}$$

$$E_{cell} = 0.34 - (-0.40).$$

$$E_{cell} = 0.74\ V$$

---

4) Iron rod immersed in $FeSO_4$ solution of 0.25M & Cu rod immersed in $CuSO_4$ solution ~~of~~ 0.45M. Standard potentials of Cu & Fe electrodes are 0.34V & −0.41 V. Give cell representation, cell reactions & calculate emf of the cell at 30°c.

A) Cell representation is

$$Fe / Fe^{2+}(0.25M) || Cu^{2+}(0.45M) / Cu$$

Cell reactions are :

At anode :- $Fe \longrightarrow Fe^{2+} + 2e^{-}$

At cathode :- $Cu^{2+} + 2e^{-} \longrightarrow Cu$

Overall reaction : $Fe + Cu^{2+} \longrightarrow Fe^{2+} + Cu$

Nernst equation for a galvanic cell at 30°C is

$$\boxed{E_{cell} = E^0_{cell} + \frac{2.303\,RT}{nF} \log \frac{[M^{n+} \text{ at cathode}]}{[M^{n+} \text{ at anode}]}}$$

$$E_{cell} = \left(E^0_{Cu^{2+}/Cu} - E^0_{Fe^{2+}/Fe}\right) + \frac{2.303\,RT}{nF} \log \frac{[Cu^{2+}]}{[Fe^{2+}]}$$

$$E_{cell} = (0.34 - (-0.41)) + \frac{2.303 \times 8.314 \times 303}{2 \times 96{,}500} \log \frac{[0.45]}{[0.25]}$$

$$E_{cell} = 0.758\,V$$

---

5) An electrochemical cell consists of iron electrode dipped in 0.1M $FeSO_4$ & silver electrode dipped in 0.05M $AgNO_3$. Write the cell representation, cell reaction & calculate the electrode potential of the cell at 298K. Given that standard reduction potential of iron & silver electrodes are −0.44 & +0.80 V.

A) Cell representation is

$$Fe / Fe^{2+}(0.1M) \,||\, Ag^{+}(0.05M) / Ag$$

Electrode reactions are

At anode :- $Fe \longrightarrow Fe^{2+} + 2e^{-}$

At cathode :- $2Ag^{+} + 2e^{-} \longrightarrow 2Ag$

Cell reaction :- $Fe + 2Ag^{+} \longrightarrow Fe^{2+} + 2Ag$

EMF of the cell

$$E^0_{cell} = E^0_{cathode} - E^0_{anode}$$

$$E^0_{cell} = E^0_{Ag^+/Ag} - E^0_{Fe/Fe^{2+}}$$

$$E^0_{cell} = 0.80 - (-0.44)$$

$$E^0_{cell} = 1.24\ V$$

Nernst equation for a Galvanic cell at 298K is

$$E_{cell} = E^0_{cell} + \frac{0.0591}{n} \log \frac{[M^{n+} \text{at cathode}]}{[M^{n+} \text{at anode}]}$$

$$E_{cell} = 1.24 + \frac{0.0591}{2} \log \frac{[0.05]}{[0.1]}$$

$$E_{cell} = 1.1927\ V$$

---

6) write the electrode reaction & calculate the electrode potential of the following cell at 298K. Given $E^0_{cell} = 1.3V$.

cell representation is

$$Cu / Cu^{2+} (1 \times 10^{-2} M) // Ag^+ (1 \times 10^{-1} M) / Ag$$

A) Electrode reactions are

At anode :- $Cu \longrightarrow Cu^{2+} + 2e^-$

At cathode :- $2Ag^+ + 2e^- \longrightarrow 2Ag$

cell reaction :- $Cu + 2Ag^+ \longrightarrow Cu^{2+} + 2Ag$

Nernst equation for a Galvanic cell at 298K is

$$E_{cell} = E^0_{cell} + \frac{0.0591}{n} \log \frac{[M^{n+} \text{at cathode}]}{[M^{n+} \text{at anode}]}$$

$$E_{cell} = 1.3 + \frac{0.0591}{2} \log \frac{[1 \times 10^{-1}]^2}{[1 \times 10^{-2}]}$$

$$E_{cell} = 1.3\ V$$